LLMBO-MO: Large Language Model-Augmented Multi-Objective Bayesian Optimization for Battery Fast-Charging Protocol Design

First A. Author, *Member, IEEE,* Second B. Author, *Fellow, OSA,* and Third C. Author, *Life Fellow, IEEE*

***Abstract*—Designing fast-charging protocols for lithium-ion batteries requires balancing charging time, thermal stress, and degradation under tight safety constraints. Bayesian optimization (BO) is attractive for this expensive black-box problem, yet its early decisions are based on few observations and typically cannot use contextual knowledge about battery operation. This paper presents LLMBO-MO, a constrained multiobjective BO framework that treats large-language-model (LLM) output as a fallible prior rather than as an objective observation or final decision. The LLM supports a diverse warm start and indicates promising regions during early optimization, while simulator evaluations alone update the Gaussian-process surrogate. Its influence on acquisition is bounded, and invalid guidance reverts to standard expected improvement. Augmented Tchebycheff scalarization handles the conflicting objectives. Under matched 56-evaluation budgets on coupled electrochemical–thermal–aging simulators, LLMBO-MO improves final hypervolume over ParEGO by 9.2% in a representative Chen2020 case and by 17.8% on average across five Ecker2015 seeds. In the latter comparison, the hypervolume standard deviation decreases from 0.0116 to 0.0024. Ablations show positive mean contributions from both LLM touchpoints, with their combination achieving the highest mean hypervolume. These results support guarded LLM guidance as a promising route to more sample-efficient Pareto search in the simulated fast-charging setting.**

***Index Terms*—Battery charging, Bayesian optimization, large language models, lithium-ion batteries, multiobjective optimization.**

I. INTRODUCTION

Lithium-ion batteries have become a key energy-storage technology for electric vehicles and other high-power applications because of their high energy density and long cycle life [1]. Shortening charging time is important for improving vehicle availability and user convenience, but aggressive charging can increase heat generation, accelerate parasitic reactions, and aggravate long-term degradation [2]. Fast-charging protocol design is therefore inherently multiobjective: charging speed must be balanced against thermal stress and battery aging without violating operational constraints.

The principal difficulty is that informative protocol evaluations are scarce. A physical cycling experiment can be slow and costly, while a high-fidelity electrochemical simulation can still limit the number of protocols evaluated within a design study. Bayesian optimization (BO) addresses this regime by combining a probabilistic surrogate with an acquisition rule that balances exploration and exploitation [3]. Attia *et al.* demonstrated closed-loop optimization of fast-charging protocols using early lifetime prediction and BO [11]. Jiang and coworkers subsequently studied minimum-time charging under operational constraints and the effect of acquisition strategies and protocol dimensionality [12], [14]. Wang and Jiang extended this line to multiple objectives through Chebyshev decomposition and constrained BO [13]. Together, these studies show that BO can reduce the evaluation burden of charging design while retaining explicit treatment of constraints and competing objectives.

For multiple conflicting objectives, a common strategy is to decompose the vector-valued problem into a sequence of scalar subproblems. ParEGO is a representative approach that combines scalarization with a probabilistic surrogate and a single-objective acquisition rule [4]. Such methods retain BO's uncertainty-aware search while covering different Pareto tradeoffs. Their earliest decisions, however, are dominated by a small initial design and generic statistical priors. Battery-specific context—for example, how charging stress varies with state of charge or why overly aggressive high-state-of-charge currents are undesirable—is not directly represented by those initial objective observations. This mismatch is most consequential near the start of optimization, when each poor query consumes a substantial fraction of the experimental budget.

Large language models (LLMs) provide a possible interface to such contextual knowledge. LLAMBO showed that LLMs can support zero-shot warm starting, surrogate modeling, and candidate sampling in data-sparse BO [10], while LGBO introduced region-level preferences that adjust a BO surrogate without changing its posterior covariance [16]. Nevertheless, an LLM response is neither a calibrated objective measurement nor a guarantee of feasibility. Directly delegating the next battery experiment to an LLM can therefore undermine the uncertainty model and constraint handling that make BO useful. The technical question is how to exploit an LLM's contextual suggestions while ensuring that evaluated data and the numerical acquisition rule remain authoritative.

To address this question, this paper proposes LLMBO-MO, an LLM-augmented multi-objective Bayesian optimization framework for lithium-ion battery fast-charging protocol design. Its central principle is to treat LLM output as a fallible prior rather than as an observation or decision rule. At initialization, validated LLM proposals form a diverse warm-

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start portfolio. During early BO iterations, weight-conditioned point or region preferences are coupled to the Gaussian-process (GP) posterior through a bounded acquisition-time mean adjustment. Only simulator outputs enter the training database; infeasible or malformed guidance has zero influence and restores ordinary expected improvement. Augmented Tchebycheff scalarization and Riesz-energy weight vectors distribute the search across the three competing objectives.

The main contributions of this paper are summarized as follows:

1) A guarded two-touchpoint architecture is formulated for constrained battery MOBO. It uses the LLM for initialization and early regional preference, while simulator observations, the GP surrogate, and the acquisition function retain control of evaluated decisions.
2) A battery-aware warm-start portfolio is combined with a weight-conditioned adaptation of region-lifted preference. Feasibility checks, bounded posterior-covariance coupling, and a fail-open path prevent invalid LLM output from becoming surrogate data or an unbounded acquisition bias.
3) Tests on Chen2020- and Ecker2015-based SPMe simulators show hypervolume gains over ParEGO under matched 56-evaluation budgets. Multi-seed and ablation results quantify variability and the separate contributions of both LLM touchpoints.

The remainder of this paper is organized as follows. Section II formulates the charging protocol optimization problem. Section III describes the electrochemical–thermal–aging simulator. Section IV presents the proposed LLMBO-MO framework. Section V reports the experimental results and ablation studies. Finally, Section VI concludes the paper.

II. Problem Formulation

A. Charging Protocol Design as CMOP

The design variable is a multi-stage constant-current charging protocol $\boldsymbol{\theta}$. The optimization problem is formulated as a constrained multi-objective optimization problem:

$$\min_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) = \{t_c(\boldsymbol{\theta}), \Delta T_p(\boldsymbol{\theta}), Q_s(\boldsymbol{\theta})\}, \tag{1}$$

where t_c is charging time, ΔT_p is the peak temperature rise above ambient, and Q_s is the capacity fade percentage. All three objectives are minimized, and the nondominated set forms the Pareto front.

B. Decision Variables and Constraints

A three-stage constant-current protocol is adopted. The decision vector is

$$\boldsymbol{\theta} = [I_1, I_2, I_3, \Delta s_1, \Delta s_2]^\top, \tag{2}$$

where I_1, I_2, I_3 are the stage currents and $\Delta s_1, \Delta s_2$ are the SOC widths of the first two stages. The third width is determined by the remaining SOC window. The bounds used in this work are listed in Table I.

The main geometric constraint is

$$\Delta s_1 + \Delta s_2 \leq 0.70, \tag{3}$$

TABLE I
Decision Variables and Bounds

Variable	Symbol	Bounds	Unit
Stage-1 current	I_1	$[2.0, 6.0]$	A
Stage-2 current	I_2	$[2.0, 5.0]$	A
Stage-3 current	I_3	$[2.0, 3.0]$	A
Stage-1 SOC width	Δs_1	$[0.10, 0.40]$	–
Stage-2 SOC width	Δs_2	$[0.10, 0.30]$	–

which ensures that a nontrivial third stage remains. During LLM candidate generation, a softer safety margin $\Delta s_1 + \Delta s_2 \leq 0.65$ is used to discourage protocols near the hard boundary.

C. Operational Constraints

During simulation, each protocol must satisfy voltage, temperature, and SOC limits:

$$V_{\text{peak}}(\boldsymbol{\theta}) \leq V_{\max}, \tag{4}$$

$$T_{\text{peak}}(\boldsymbol{\theta}) \leq T_{\max}, \tag{5}$$

$$0 \leq \text{SOC}(t) \leq 0.80. \tag{6}$$

Protocols that violate hard operational constraints receive penalty objective values and are not allowed to dominate feasible protocols.

III. Electrochemical–Thermal Evaluation Model

The proposed optimization framework requires a forward model that maps each charging protocol to its charging-time, thermal, and degradation responses. Since the focus of this work is the optimization strategy rather than the development of a new battery model, all methods are evaluated using the same physics-based simulation environment. The simulator combines an electrochemical Single Particle Model with electrolyte dynamics (SPMe), a lumped thermal model, and a protocol-level semi-empirical degradation proxy. This construction retains the principal electrochemical and thermal effects relevant to fast charging while remaining sufficiently efficient for repeated black-box evaluations.

A. Electrochemical–Thermal Dynamics

The electrochemical dynamics are represented by the SPMe, a reduced-order approximation of the Doyle–Fuller–Newman model that retains electrolyte concentration dynamics while representing each porous electrode by a characteristic spherical active-material particle [6]. Solid-phase lithium diffusion, electrolyte transport, and interfacial reaction kinetics are therefore retained at a substantially lower computational cost than a full-order porous-electrode model. The numerical implementation is based on PyBaMM [7].

The electrochemical model is coupled to a spatially lumped thermal balance, which may be written in the form

$$C_{\text{th}} \frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{gen}} - hA\left(T - T_{\text{amb}}\right), \tag{7}$$

where C_{th} is the effective thermal capacitance of the cell, $\dot{Q}_{\text{gen}}$ denotes the heat generated by electrochemical reactions, ohmic losses, and reversible effects, h is the effective heat transfer coefficient, and A is the heat-transfer area. The thermal objective is summarized by the peak temperature rise

$$\Delta T_p(\boldsymbol{\theta}) = \max_t T(t) - T_{\text{init}}. \tag{8}$$

The Chen2020 parameter set is used for the primary LG INR21700-M50 study [8]. To examine whether the optimization behavior persists under a different cell parameterization, additional experiments are conducted with the Ecker2015 parameter set [9]. Cell-dependent voltage and temperature limits are specified in the experimental setup, while the optimization procedure itself is kept unchanged.

B. Three-Stage Charging Realization

For the decision vector $\boldsymbol{\theta} = [I_1, I_2, I_3, \Delta s_1, \Delta s_2]^\top$, the charging interval from SOC 0 to 0.8 is partitioned into three constant-current stages. The third SOC width is determined by

$$\Delta s_3 = 0.8 - \Delta s_1 - \Delta s_2. \tag{9}$$

The stages are simulated sequentially, with the terminal electrochemical and thermal states of one stage serving as the initial states of the next.

For a stage spanning Δs_k, the nominal charging duration follows from charge balance,

$$t_k = \frac{Q_{\text{eff}} \Delta s_k}{I_k^{\text{cell}}} 3600, \qquad k = 1, 2, 3, \tag{10}$$

where Q_{eff} is the effective cell capacity in Ah and I_k^{cell} is the physical charging current in A. The total charging time is therefore

$$t_c(\boldsymbol{\theta}) = \sum_{k=1}^{3} t_k. \tag{11}$$

For the 5-Ah Chen2020 case, the protocol-current coordinates coincide with the corresponding physical currents. For other cell parameterizations, the same protocol coordinates are capacity-scaled before simulation so that comparable charging-rate regimes are explored rather than identical absolute currents.

During each simulation, the voltage and temperature trajectories are monitored. A protocol is regarded as feasible only when the prescribed SOC interval is completed without violating the cell-specific voltage and temperature limits. Infeasible evaluations are excluded from the feasible Pareto set used for performance assessment.

C. Protocol-Level Degradation Proxy

A single fast-charging simulation is insufficient to resolve long-horizon capacity fade directly without introducing a substantially more expensive aging model. Accordingly, the third objective is defined as a semi-empirical protocol-level degradation proxy. Let $\bar{s}$, $\bar{T}$, and $\bar{I}$ denote the trajectory-averaged SOC, temperature, and charging current, respectively. An empirical effective-capacity function, $\text{Cap}(\bar{s}, \bar{T}, \bar{I})$, is used to summarize the combined influence of these operating conditions. The resulting degradation indicator is

$$Q_s(\boldsymbol{\theta}) = 100 \, \frac{Q_{\text{eff}}}{\text{Cap}(\bar{s}, \bar{T}, \bar{I})}. \tag{12}$$

A smaller Q_s corresponds to a less severe charging condition under the adopted empirical model. Importantly, Q_s is used as a *relative degradation indicator* for comparing charging protocols under a common simulation environment; it should not be interpreted as a chemistry-specific prediction of the absolute capacity loss caused by a single charging event. This distinction is maintained throughout the experimental analysis.

D. Black-Box Evaluation Interface

The coupled simulator defines the mapping

$$\mathcal{S} : \boldsymbol{\theta} \longmapsto [t_c(\boldsymbol{\theta}), \Delta T_p(\boldsymbol{\theta}), Q_s(\boldsymbol{\theta})]^\top, \tag{13}$$

together with a feasibility indicator. From the perspective of Bayesian optimization, this simulator is treated as an expensive black box: the optimizer observes only previously evaluated protocols, their objective values, and feasibility information, rather than the internal electrochemical states used to generate them. Consequently, the same optimization framework can be evaluated under different battery parameterizations without changing its decision logic.

IV. Proposed LLMBO-MO Framework

A. Framework Overview

LLMBO-MO is designed to incorporate battery-domain knowledge without replacing the numerical decision mechanism of Bayesian optimization (BO). The LLM is used at two controlled touchpoints: it first proposes a pool of physically plausible initial protocols, and, during the early BO iterations, it expresses a coarse preference for a promising point or region. All objective values entering the surrogate are obtained from the simulator; unevaluated LLM suggestions are never inserted as observations. The next protocol is selected by scalarization, a Gaussian-process (GP) surrogate, and expected improvement (EI), followed by simulator evaluation and Pareto-set update.

At iteration t, a weight vector $\boldsymbol{w}^{(t)}$ specifies the current objective trade-off. The evaluated multi-objective database is re-scalarized under $\boldsymbol{w}^{(t)}$, a GP is fitted to the resulting scalar targets, and the LLM may provide a point/region preference from the current optimization state. Following the region-lifted preference principle of LGBO [16], the valid preference is converted into a mean adjustment used only during acquisition. In the configuration evaluated in Section V, this adjustment uses GP posterior cross-covariance; no separate LLM acquisition bonus, LLM reranking, or direct insertion of LLM-generated iteration candidates is used. If a preference cannot be converted into a valid feasible region, the method fails open to ordinary EI.

Figure 1 summarizes the constrained warm start, the numerical MOBO loop, and the two advisory LLM touchpoints while keeping the simulator-evaluated data path explicit.

B. Objective Transformation and Scalar Decomposition

The three objectives in (1) have substantially different numerical scales. We therefore transform charging time and

Fig. 1. Unified LLMBO-MO workflow. *A:* constrained warm-start construction combines domain-informed LLM candidates with physics priors and space-filling samples, then repairs, screens, and verifies the portfolio in the simulator. *B:* the numerical multi-objective BO loop performs scalarization, GP fitting, expected-improvement optimization, and simulator evaluation. The LLM enters only through the two advisory touchpoints; invalid or infeasible iteration-level guidance falls back to plain EI, and only simulator outputs update the optimization state.

capacity fade logarithmically, while retaining the temperature-rise objective in its original scale:

$$\tilde{\mathbf{f}}(\boldsymbol{\theta}) = \left[\log_{10} t_c(\boldsymbol{\theta}) \quad \Delta T_p(\boldsymbol{\theta}) \quad \log_{10} Q_s(\boldsymbol{\theta})\right]^\top . \tag{14}$$

Let $\tilde{z}_i^*$ denote the corresponding transformed ideal value. For objective i, the ideal-point gap is scaled as

$$\bar{f}_i(\boldsymbol{\theta}) = \frac{\left|\tilde{f}_i(\boldsymbol{\theta}) - \tilde{z}_i^*\right|}{s_i}, \qquad s_i = \max(y_{\max,i} - y_{\min,i}, \epsilon_i)\,, \tag{15}$$

where $y_{\min,i}$ and $y_{\max,i}$ are computed from the transformed feasible observations. The small range floor ϵ_i prevents unstable scaling when the early observations occupy only a narrow objective interval.

Following the decomposition strategy of ParEGO [4], the vector-valued problem is converted at each iteration into an augmented Tchebycheff subproblem:

$$g^{\text{tch}}(\boldsymbol{\theta} \mid \boldsymbol{w}) = \max_{i\in\{1,2,3\}} \left\{w_i \bar{f}_i(\boldsymbol{\theta})\right\} + \eta \sum_{i=1}^{3} w_i \bar{f}_i(\boldsymbol{\theta}), \tag{16}$$

where $\boldsymbol{w} \in \mathbb{R}^3_+$, $\sum_i w_i = 1$, and $\eta = 0.05$. The augmentation term reduces ties around the maximum weighted deviation.

To distribute the scalar subproblems over distinct trade-off directions, a Riesz s-energy reference set is used [15]. For a weight set $\mathcal{W} = \{\boldsymbol{w}^{(1)}, \ldots, \boldsymbol{w}^{(N)}\}$ on the simplex, its energy is

$$E_s(\mathcal{W}) = \sum_{p<q} \left\|\boldsymbol{w}^{(p)} - \boldsymbol{w}^{(q)}\right\|^{-s}. \tag{17}$$

The implementation initializes $\mathcal{W}$ from a Das–Dennis simplex lattice and applies projected repulsive-gradient relaxation to reduce (17), rather than assuming that the global minimum of the energy is available in closed form. Weight vectors are then visited without replacement before the set is reshuffled, which provides dispersed coverage of the three-objective trade-off space.

C. Gaussian-Process Surrogate

For the current weight vector, a single-output GP is fitted to $\{(\boldsymbol{\theta}_j, g_j^{\mathrm{tch}})\}_{j=1}^{|\mathcal{D}|}$. Each decision variable is first mapped to $[0,1]$. The covariance model is an automatic-relevance-determination Matérn-5/2 kernel with a white-noise term,

$$k(\boldsymbol{\theta},\boldsymbol{\theta}') = \sigma_f^2\left(1+\sqrt{5}r+\frac{5r^2}{3}\right)e^{-\sqrt{5}r} + \sigma_n^2\delta_{\boldsymbol{\theta}\boldsymbol{\theta}'}, \qquad (18)$$
$$r^2 = \sum_{d=1}^{5}\frac{(x_d - x_d')^2}{\ell_d^2},$$

where x_d denotes the normalized coordinate and ℓ_d is its learned length scale. Kernel hyperparameters are estimated by marginal-likelihood optimization with multiple restarts. The scalar targets are standardized during GP fitting. The posterior mean $\mu_t(\boldsymbol{\theta})$, standard deviation $\sigma_t(\boldsymbol{\theta})$, and posterior covariance are subsequently used by EI and the region-lift mechanism.

D. LLM Integration Through Two Controlled Touchpoints

1) Touchpoint 1: Constrained Warm-Start Portfolio: The warm-start prompt provides the cell/model context, decision-variable bounds, the hard constraint in (3), and battery-design heuristics such as avoiding unnecessarily aggressive high-SOC charging and considering decreasing current profiles. The LLM overgenerates a candidate pool. Candidate responses are parsed and repaired, after which nonfinite or malformed points, out-of-bound protocols, hard $\Delta s_1 + \Delta s_2$ violations, and duplicates are removed.

The remaining candidates are selected greedily to balance the LLM's reported confidence, distance from the soft $\Delta s_1 + \Delta s_2$ limit, monotone-current structure, and geometric diversity. For candidate $\boldsymbol{\theta}$, define

$$p_{\mathrm{soft}}(\boldsymbol{\theta}) = \frac{[\Delta s_1 + \Delta s_2 - s_{\mathrm{soft}}]_+}{s_{\mathrm{hard}} - s_{\mathrm{soft}}}, \qquad (19)$$

with $s_{\mathrm{soft}} = 0.65$ and $s_{\mathrm{hard}} = 0.70$, and let $\mathbb{I}_{\mathrm{mono}}(\boldsymbol{\theta}) = 1$ when $I_1 \geq I_2 \geq I_3$. The score used when extending a selected portfolio $\mathcal{C}_{\mathrm{sel}}$ is

$$J_{\mathrm{ws}}(\boldsymbol{\theta}) = c_{\mathrm{ws}}(\boldsymbol{\theta}) - \omega_p p_{\mathrm{soft}}(\boldsymbol{\theta}) + \omega_m \mathbb{I}_{\mathrm{mono}}(\boldsymbol{\theta}) + \omega_d d_{\min}(\boldsymbol{\theta}, \mathcal{C}_{\mathrm{sel}}), \qquad (20)$$

where $c_{\mathrm{ws}} \in [0,1]$ is the candidate confidence and $d_{\min}$ is the minimum Euclidean distance in normalized decision space to an already selected protocol. A small number of near-boundary probes can be retained to avoid collapsing the initial design into only conservative schedules. If too few valid LLM candidates remain, deterministic battery heuristics and Latin-hypercube/random samples complete the initialization. Only after simulator evaluation are these protocols added to $\mathcal{D}$.

The complete hypothesis-to-observation path is shown in the constrained warm-start block of Fig. 1.

2) Touchpoint 2: Weight-Conditioned Point/Region Preference: During an early guidance window, the LLM receives the current weight vector $\boldsymbol{w}^{(t)}$, ideal point and objective scaling, the best scalarized observations, recent evaluated protocols, short-horizon hypervolume feedback, boundary-failure statistics, and feedback on the previously adopted region. The prompt explicitly asks for a preference rather than the next experiment. The output is restricted to either a raw-coordinate point $\boldsymbol{\theta}_{\mathrm{pref}}$, a hyper-rectangular region $[\mathbf{lb}, \mathbf{ub}]$, or no preference, together with a confidence $c_t \in [0,1]$ and a short mechanistic rationale.

The interface checks the output format, coordinate system, direction of the preference, and parameter bounds. A region is clipped/repaired to admissible width and volume limits and to the Δs safety geometry. A point preference is converted to a small local box before the same processing. Low confidence does not by itself create a hard rejection in the reported region-lift variant; instead, c_t directly scales the strength of the lift. If the response is absent, unparsable, structurally invalid, or yields no feasible anchors, its influence is set to zero and acquisition reduces to ordinary EI.

E. Posterior-Covariance Region Lift for Battery MOBO

The general region-lift idea follows LGBO [16]; here it is specialized to a sequentially scalarized, constrained battery-charging problem and to a posterior-cross-covariance coupling. Let a valid preferred region be represented by M feasible Sobol anchors $\mathcal{G} = \{\mathbf{g}_1, \ldots, \mathbf{g}_M\}$. The reported configuration uses uniform anchor weights,

$$\mathbf{v} = \frac{1}{M}\mathbf{1}_M. \qquad (21)$$

All quantities in the following lift are expressed in the standardized GP target space. Let $\boldsymbol{\Sigma}_{GG,t}$ be the posterior covariance matrix among the anchors. The posterior variance of the regional average is

$$V_{G,t} = \mathbf{v}^\top \boldsymbol{\Sigma}_{GG,t}\mathbf{v}. \qquad (22)$$

The confidence-normalized lift strength is

$$\lambda_t = \frac{c_t}{\sqrt{\max(V_{G,t}, \varepsilon_v)}}, \qquad (23)$$

where $\varepsilon_v > 0$ prevents numerical division by a vanishing regional variance. For any candidate $\boldsymbol{\theta}$, let $\boldsymbol{\Sigma}_{\theta G,t}$ denote the GP posterior cross-covariance between $\boldsymbol{\theta}$ and the anchors. The acquisition-time mean becomes

$$\mu_t^{\mathrm{L}}(\boldsymbol{\theta}) = \mu_t(\boldsymbol{\theta}) - \lambda_t \boldsymbol{\Sigma}_{\theta G,t}\mathbf{v}, \qquad (24)$$

while the predictive variance is unchanged,

$$\left(\sigma_t^{\mathrm{L}}(\boldsymbol{\theta})\right)^2 = \sigma_t^2(\boldsymbol{\theta}). \qquad (25)$$

For this minimization problem, the negative mean shift makes points that are posterior-correlated with the preferred region more competitive under EI. The operation is an acquisition-time preference transformation only: it neither modifies the training targets nor treats LLM statements as pseudo-observations. In the reported configuration, the lift is active only during the first $T_{\mathrm{L}} = 16$ BO iterations; subsequent iterations use the unmodified GP posterior.

The numerical-loop block of Fig. 1 shows how this optional advisory branch joins the otherwise conventional MOBO loop without changing the observation database or GP variance.

F. Expected Improvement and Stagnation-Aware Search

Let $g_{\min,t}$ be the best scalarized value observed for the current weight vector. EI is evaluated using the lifted mean when a valid coupling is active:

$$z_t(\boldsymbol{\theta}) = \frac{g_{\min,t} - \mu_t^{\mathrm{L}}(\boldsymbol{\theta})}{\sigma_t(\boldsymbol{\theta})}, \tag{26}$$

$$\alpha_{\mathrm{EI}}(\boldsymbol{\theta}) = \left(g_{\min,t} - \mu_t^{\mathrm{L}}(\boldsymbol{\theta})\right)\Phi(z_t) + \sigma_t(\boldsymbol{\theta})\phi(z_t). \tag{27}$$

When no valid lift is available, $\mu_t^{\mathrm{L}} = \mu_t$, giving standard EI. The next protocol is selected as

$$\boldsymbol{\theta}_{t+1} = \arg\max_{\boldsymbol{\theta}\in\Theta_{\mathrm{feas}}} \alpha_{\mathrm{EI}}(\boldsymbol{\theta}). \tag{28}$$

Acquisition optimization uses a mixed candidate pool containing the current best protocol, Gaussian and uniform samples, and multi-start L-BFGS-B solutions of EI. Each proposed point is repaired to satisfy the decision bounds and the hard $\Delta s_1 + \Delta s_2$ constraint. Stagnation does not add an LLM-specific bonus to (27); instead, it expands the Gaussian restart dispersion, thereby increasing exploration while preserving EI as the ranking criterion. If EI is numerically flat over the candidate pool, the implementation falls back to the point with largest predictive uncertainty.

G. Overall Algorithm

Algorithm 1 LLMBO-MO

Require: Simulator $\mathcal{S}$, LLM $\mathcal{L}$, BO budget T, warm-start count n_{ws}, weight set $\mathcal{W}$, guidance horizon T_{L}
Ensure: Nondominated set $\mathcal{P}$ and hypervolume trace

```
Generate or load the Riesz-relaxed weight set W
Query L for an overgenerated warm-start pool
Repair/filter the pool and select n_ws protocols by J_ws
for each selected warm-start protocol do
    Evaluate S and add the resulting observation to D
end for
for t = 1 to T do
    Select w^(t) ∈ W and update transformed objective scaling
    Compute g^tch(· | w^(t)) for all feasible observations
    Fit the GP to the current scalarized database
    Set the region-lift coupling to inactive
    if t ≤ T_L then
        Query L for a point/region preference conditioned on w^(t)
        Repair the region and construct feasible Sobol anchors
        if the preference and regional covariance are valid then
            Build λ_t and the posterior-covariance mean lift in (24)
        end if
    end if
    Maximize EI using the lifted mean if active, otherwise plain EI
    Evaluate S(θ_{t+1}) and update D, P, and HV
end for
return P
```

V. Experiments

A. Experimental Protocol

All results are obtained in simulation; no new physical-cell measurements are reported. The simulator uses the SPMe formulation in Section III with either the Chen2020 or Ecker2015 parameter set. The optimizer minimizes charging time, peak temperature rise, and capacity fade. Unless noted otherwise, one run consists of six initialization evaluations followed by 50 BO iterations, giving 56 simulator evaluations.

The archive field `canonical_hv` is used as the primary quality measure. It is the hypervolume of the nondominated set in the transformed objective space under a fixed benchmark-specific reference box. The value is dimensionless but not restricted to $[0, 1]$; absolute values are therefore compared only within the same cell parameterization. Reported uncertainties are sample standard deviations across seeds.

The figures use three distinct scopes. Fixed-budget tables contain either the Chen2020 seed-8409 case or five paired Ecker2015 runs. The Pareto-count curve aggregates five archived runs per method, whereas the three-dimensional Pareto plot pools Chen2020 LLMBO-MO archives to expose the attainable design envelope. These scopes are stated in the captions and are not combined into one statistical estimate.

B. Baselines

ParEGO is the principal BO baseline because it uses the same scalarized surrogate-and-acquisition structure without language-model guidance. NSGA-II provides an evolutionary reference, while DISK and PIMD provide additional population-based comparators in the archived Chen2020 plot. All methods are evaluated through simulator calls; LLM suggestions are never counted as objective observations before simulation.

C. Fixed-Budget Search Behavior

Figure 2 separates final-set quality from archive growth. The left panel retains the original canonical-HV scale, which makes the early jumps and subsequent plateaus visible. The right panel counts nondominated protocols after each BO iteration; it is an archive diagnostic, not an evaluation-normalized quality metric.

Table II reports the matched fixed-budget values used for the numerical comparison. On the Chen2020 seed-8409 case, LLMBO-MO increases canonical HV from 0.3523 to 0.3848. This is a single-seed result and is not treated as evidence of a population-level mean improvement. On Ecker2015, LLMBO-MO improves the five-seed mean from 1.5866 to 1.8684, an absolute gain of 0.2818 (17.8%). Its sample standard deviation is reduced by approximately 79% relative to ParEGO.

TABLE II
FIXED-BUDGET RESULTS AT 56 SIMULATOR EVALUATIONS

Cell model	Method	Runs	Final canonical HV
Chen2020	ParEGO reference	1	0.3523
	LLMBO-MO	1	**0.3848**
Ecker2015	ParEGO	5	1.5866 ± 0.0116
	LLMBO-MO	5	**1.8684 ± 0.0024**

The archive-size panel in Fig. 2 ends at 49.4 ± 2.4 nondominated protocols for LLMBO-MO and 45.8 ± 0.7 for ParEGO. Because the archived LLMBO-MO runs begin from a larger

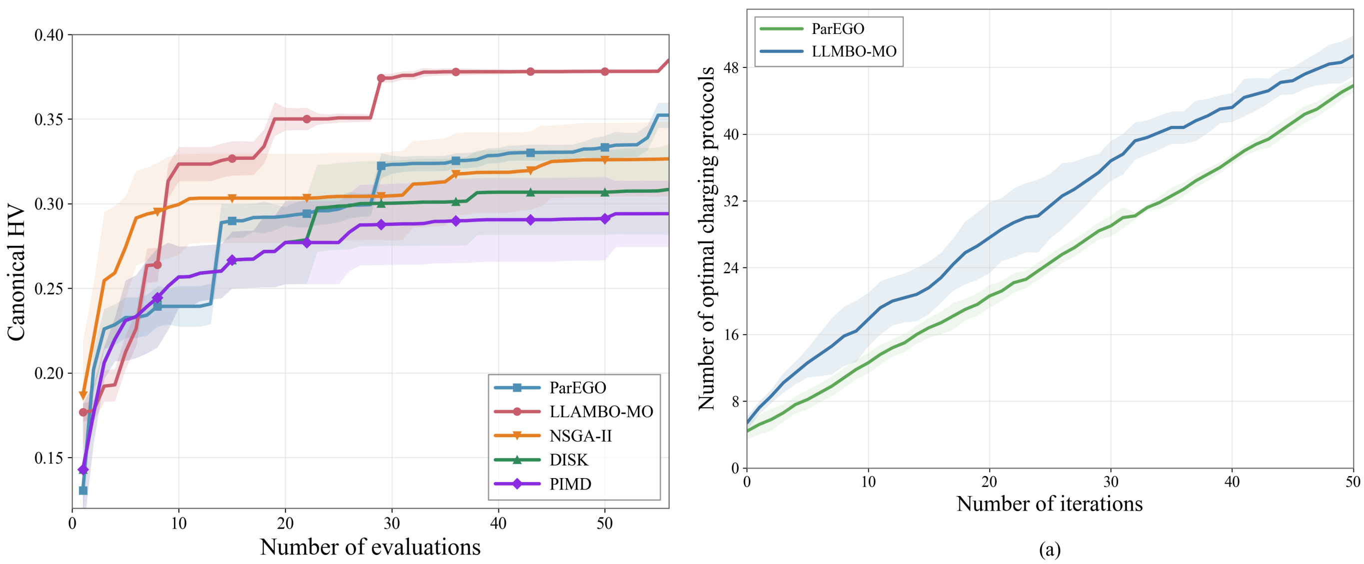


Fig. 2. Chen2020 search diagnostics. *Left:* representative canonical-HV trajectories. The LLMBO-MO and ParEGO centerlines are from seed 8409; NSGA-II, DISK, and PIMD are five-seed means. The LLMBO-MO and ParEGO ribbons are archived proxy dispersion envelopes and are not confidence intervals; the remaining ribbons show one sample standard deviation. *Right:* five-seed mean number of nondominated charging protocols, with ± 1 sample-standard-deviation bands. The underlying archives contain different initial database sizes, so the right panel is used only to describe archive growth.

evaluated database, this difference is not used to claim better sample efficiency. The fixed-budget HV values in Table II remain the primary comparison.

D. Pareto Structure and Charging Profiles

Figure 3 connects the objective-space trade-offs to time-domain charging behavior. The left panel pools 730 evaluated Chen2020 points from the archived LLMBO-MO runs and displays the 233 nondominated points. Pooling is used only to reveal the design envelope; the resulting cloud is not used to compute Table II. The objective values of the starred solutions are listed directly beneath the plot.

Point A occupies the fast-charging edge and incurs the largest temperature rise and capacity fade. Point B minimizes fade among the five marked solutions, whereas C lies near the knee of the time–temperature trade-off. Points D and E reduce thermal stress to approximately 1.53 K at the cost of longer charging. The replayed tuned protocol in the right panel uses a brief 5-A middle stage, shortens charging relative to the warm-start and baseline profiles, and correspondingly produces a larger temperature excursion. All three voltage traces remain below the 4.3-V limit.

E. Ablation Study

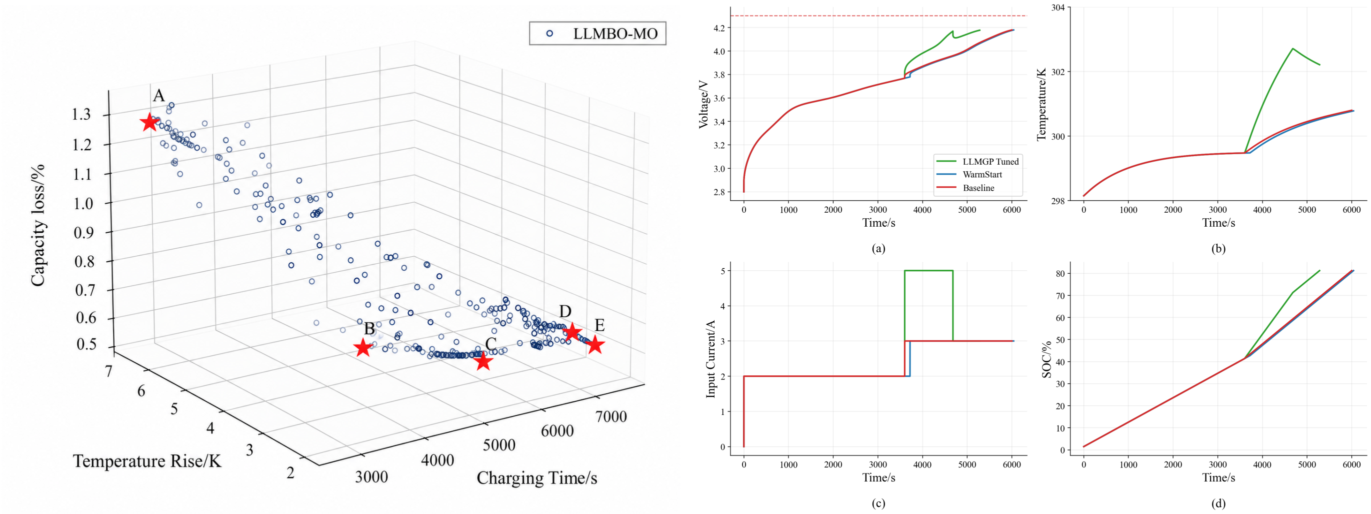
The four-group summary isolates the two LLM touch-points at the configuration level. Baseline, WarmStart, and LLM_Region are taken from the May 22 five-seed archive; the full method is taken from the paired May 23 rerun using the same selected warm-start initialization. Thus, Fig. 4 preserves the five-seed distributions but is a consolidated archive summary, not a single simultaneously executed four-arm batch.

Warm starting produces the larger and more repeatable individual-component gain, improving the mean by 0.0066 and winning four of five paired seeds. Region guidance also improves the mean, but with greater dispersion. Their combination gives the highest mean HV, 0.3932, although the 3/5 win count and the consolidated-batch construction caution against attributing the gain to a uniform per-seed interaction.

F. Runtime and Scope

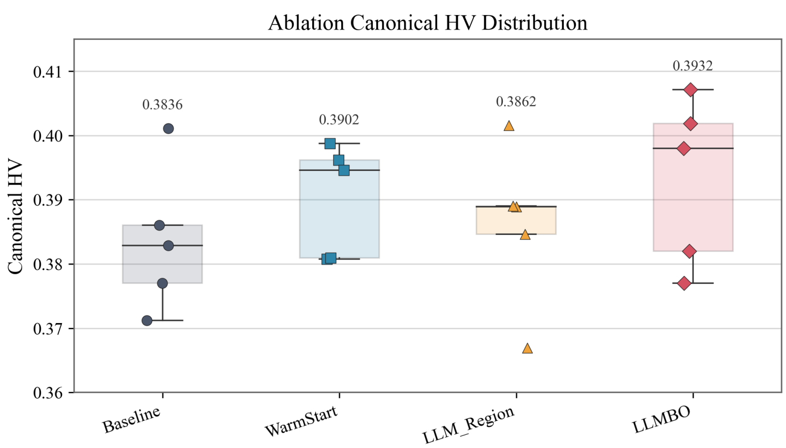
The archived Chen2020 timing runs give 194.3 ± 11.8 s for NSGA-II, 252.4 ± 17.7 s for ParEGO, and 440.3 ± 31.3 s for LLMBO-MO. LLMBO-MO is approximately 1.74 times slower than ParEGO because wall time includes LLM requests, response validation, region processing, and a larger acquisition pool. Hardware and API-latency metadata were not retained, so these values document overhead within the archive rather than a portable runtime benchmark.

The evidence is limited to simulation, two parameterizations, and at most five seeds. The Chen2020 comparison figure also mixes single-seed centerlines with multi-seed baselines, as stated in its caption. Physical-cell validation and a fully paired multi-seed study with identical initial databases remain necessary before making claims about hardware-level charging performance or general statistical superiority.



Point	Charging time (s)	Temp. rise (K)	Fade (%)
A	2880	7.567	1.261
B	5295	4.996	**0.538**
C	6151	2.837	0.571
D	6928	1.535	0.685
E	7199	**1.528**	0.640

Fig. 3. Objective-space and time-domain views of Chen2020 charging protocols. *Left:* pooled LLMBO-MO nondominated set; stars A–E mark representative trade-offs from fast/high-stress to slow/low-stress charging. *Right:* simulator replay of one tuned LLMBO-MO protocol, a warm-start-only protocol, and the numerical baseline: (a) terminal voltage, (b) cell temperature, (c) applied current, and (d) state of charge. The archived legend "LLMGP Tuned" denotes the tuned LLMBO-MO configuration. These curves are simulated trajectories rather than physical-cell measurements.



FIVE-SEED NUMERICAL SUMMARY

Variant	Final HV	Δ	Wins
Baseline	0.3836 ± 0.0101	–	–
WarmStart	0.3902 ± 0.0078	+0.0066	4/5
LLM_Region	0.3862 ± 0.0112	+0.0026	4/5
Full LLMBO	**0.3932** ± 0.0117	**+0.0096**	3/5

Fig. 4. Chen2020 four-group ablation over five seeds and 50 BO iterations. *Left:* final canonical-HV distributions; markers show individual seeds, boxes show the interquartile range and median, and annotations give the means. *Right:* the corresponding mean, sample standard deviation, gain over the baseline, and paired-seed win count. The full method combines constrained warm starting and iteration-level LLM region guidance.

VI. Conclusion

This paper presented LLMBO-MO, a large language model-augmented multi-objective Bayesian optimization framework for battery fast-charging protocol design. The framework uses the LLM at two guarded touchpoints: warmstart generation and iteration-level point/region guidance. The optimizer keeps the final decision inside a conventional BO loop through Riesz/Tchebycheff scalarization, GP modeling, EI acquisition, risk penalties, and simulator feedback. Experiments on SPMe-based simulators show that LLMBO-MO improves hypervolume over ParEGO by 9.2% on a Chen2020 case and by 17.8% across five Ecker2015 seeds, with reduced seed-to-seed variation. An ablation study confirms that both warmstart and iteration-level guidance contribute positively, and their combination achieves the highest mean hypervolume.

Future work will add stronger statistical validation across more cell models, study locally hosted LLM backends to reduce runtime overhead, and extend the framework to variable-stage protocols with mixed discrete-continuous decision spaces.

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